

The Science Behind CPTG

A Geometric Alternative to Dark Matter, Dark Energy, and MOND

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Abstract

Curvature Polarization Transport Gravity (CPTG) differs from dark-halo, dark-energy, and interpolation-law approaches because it does not begin by adding an unseen sector where gravitational support appears to be missing. It begins with a native geometric theory: baryon-sourced curvature polarization, curvature transport, geometric branch structure, active response, and comparison-coordinate projection. Its formulas are derived from the core framework, locked before testing, and scaled into the observational language required by each data product. Galaxy rotation curves, cluster lensing maps, CMB likelihood coordinates, BBN abundance tables, lithium source-network channels, Hubble comparison branches, DESI ruler coordinates, and high-redshift stress tests are different observational interfaces, but in CPTG they are reduced expressions or scalable projections of the same geometric source.

This paper states the leading difference between CPTG and competing frameworks. Standard dark-matter practice treats missing gravitational behavior as evidence for a movable non-baryonic component; MOND-style theories introduce low-acceleration laws; and modern cosmological fits rely on parameter estimation, nuisance layers, and calibration machinery. CPTG instead asks whether a compact locked geometric rule set can reproduce structure, lensing, abundance, CMB, distance, and growth behavior without sector-by-sector tuning. Capabilities such as Structural Mode N , comparison coordinates, active-gate cluster response, the Hubble-tension bridge, the lithium gate, CMB map and likelihood bridges, high-redshift stress tests, compact-object branches, and vanishing-star collapse interpretations arise from one scalable geometric framework rather than from independent adjustable fixes.

1 Introduction to CPTG

The question is simple:

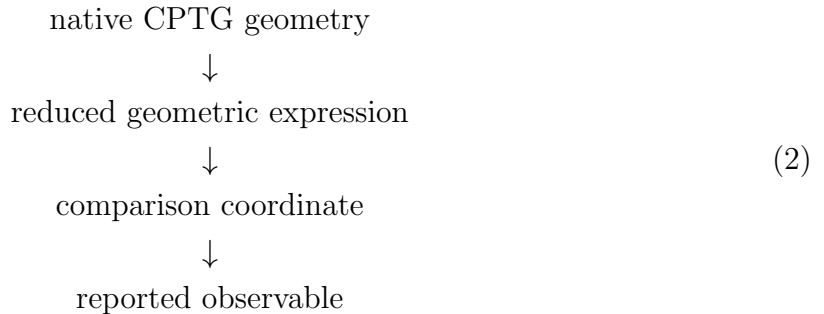
What makes CPTG different from competing theories? (1)

The answer is not merely that CPTG [6] can be compared with many data sectors. The answer is that CPTG carries one derived geometric framework across those sectors

without changing the formulas to suit each target. The observational interfaces differ because the measurements differ. The native source does not .

A dark-halo model can add or redistribute unseen mass where gravitational support is needed. A parameter-fit cosmology can move nuisance layers, calibration choices, or sector-specific parameters inside a likelihood analysis. A phenomenological acceleration-law theory can choose or adjust an interpolation rule. CPTG takes a different path. It applies derived geometry first and lets the observational comparison stand or fail.

The core validation sequence is



This order matters. The comparison coordinate is not the theory. It is the observational language required by a particular data product, map, likelihood, abundance table, or compressed summary. The theory is the native curvature-polarization and curvature-transport structure from which the reduced formula is derived.

2 Why the Dark-Matter Assumption Is Operationally Fragile

The observations that motivate dark matter are real. Galaxy rotation curves, cluster lensing, large-scale structure, CMB acoustic structure, and gravitational growth all require an explanation [2, 4, 5]. The fragile part is not the observation. The fragile part is the operational assumption that the missing gravitational response must be an unseen material component that can be assigned, shaped, profiled, suppressed, or redistributed wherever the model requires support.

In practice, dark matter functions as a flexible inference layer. In galaxies, halo profiles can be adjusted around the baryonic source. In clusters, collisionless mass can be placed where lensing demands it. In simulations, baryonic feedback, substructure, concentration, merger history, and selection effects can be modified to recover a target. In cosmology, additional calibration and nuisance machinery is used to reconcile data products that do not naturally report the same preferred coordinate.

CPTG does not deny the gravitational anomalies. It challenges the interpretation of their source. The CPTG alternative is that the missing-gravity signal is a nonlinear geometric response of spacetime to baryonic structure. This makes the question sharper:

$$\text{Is the anomaly missing matter, or is it missing geometry?} \tag{3}$$

3 The CPTG Starting Point

CPTG begins with baryon-sourced curvature polarization and curvature transport [6]. In schematic form, the nonlinear field structure may be written as

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (4)$$

Here $\Pi_{\mu\nu}^{\text{pol}}$ and $\Theta_{\mu\nu}^{\text{trans}}$ are not dark fluids, halo components, or empirical correction terms. They represent the curvature-polarization and curvature-transport sectors of the geometry. The operational hierarchy is

$$\begin{array}{c} \text{nonlinear CPTG field structure} \\ \downarrow \\ \text{symmetry and scale reduction} \\ \downarrow \\ \text{scalar closure law} \\ \downarrow \\ \text{observed comparison coordinate} \end{array} \quad (5)$$

The scalar laws are therefore not unrelated postulates. They are reduced expressions of the same geometric field structure under the symmetry appropriate to the problem being tested.

4 The Compact Geometric Branch

The current geometric branch is organized by the locked quantities

$$p_C = \pi, \quad (6)$$

$$p_{\text{ac}} = 3 - \frac{\pi}{100}, \quad (7)$$

$$G_T = \frac{p_{\text{ac}}}{p_C}, \quad (8)$$

and the native acceleration anchor

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} \text{ m s}^{-2}. \quad (9)$$

The late/geometric and transport branch weights are

$$B_\pi = 1 - \frac{1}{\pi}, \quad (10)$$

$$T_\pi = \frac{1}{\pi}. \quad (11)$$

The corresponding branch expansion law is

$$E_\pi(a)^2 = B_\pi + T_\pi a^{-p_C}. \quad (12)$$

These quantities are not posterior fit products. They are branch definitions and transport weights. Once fixed, the observational work becomes coordinate expression: how the same branch appears in supernova, CMB, BBN, weak-lensing, BAO, DESI, and clock variables [7, 8].

5 What Competing Theories Usually Do

Competing frameworks are not all wrong in the same way. They differ in mechanism and validation style. The important point is that CPTG does not belong to the same class as adjustable dark-sector placement or interpolation-law fitting [1].

Framework	Primary explanatory device	Typical operational freedom	CPTG difference
Dark-halo modeling	Non-baryonic mass distribution assigned around visible matter	Halo profile, concentration, substructure, feedback, merger history, and baryonic prescriptions	CPTG does not assign a hidden mass profile; it solves or projects curvature response from baryonic geometry.
Λ CDM cosmology	Dark matter, dark energy, fitted cosmological parameters, nuisance layers	Global parameter estimation, calibration choices, likelihood profiling, sector-specific nuisance parameters	CPTG uses fixed geometric branch quantities and translates them into comparison coordinates.
MOND-like theories	Modified low-acceleration relation	Interpolation functions, acceleration-law choices, mass-to-light freedom in many implementations	CPTG obtains low-acceleration behavior as a curvature-polarization reduction and also predicts structural organization.
CPTG	Baryon-sourced curvature polarization, curvature transport, scalable geometric projection	Formula changes are not allowed after branch lock; observational coordinates differ but the native source remains fixed	CPTG tests whether one geometric branch can scale across independent observables without sector-by-sector tuning.

Table 1: Comparison between CPTG and common competing theory classes.

6 What Curvature Polarization Means

In CPTG, curvature polarization is the local nonlinear response of spacetime curvature to baryonic structure. It is not a dark matter density, a hidden halo profile, or a freely assigned correction term. It is the way the geometric field reorganizes around ordinary matter when the field becomes weak, extended, or structurally organized.

The simplest way to describe the idea is by analogy with polarization in a material medium. In electromagnetism, a medium can respond to a source by developing an induced polarization field. In CPTG, baryonic matter sources a curvature response. The observed gravitational field is therefore not only the direct baryonic field. It also includes the curvature-polarization response generated by that baryonic structure.

The hierarchy is

$$\begin{array}{ccc}
& \text{baryonic source geometry} & \\
& \downarrow & \\
& \text{curvature-polarization response} & (13) \\
& \downarrow & \\
& \text{observed gravitational support} &
\end{array}$$

In the galaxy reduction, this appears as a solved acceleration response rather than an assigned halo. The observed acceleration is not obtained by adding an arbitrary dark component. It is obtained by solving the reduced curvature-polarization equation sourced by the observed gas, disk, and bulge fields.

A representative reduced form is

This expression is schematic. It separates the direct baryonic field, the polarization response, and the geometric transport contribution; the full reduced galaxy equation includes the nonlinear suppression structure shown later in the MOND comparison section.

$$g_{\text{CPTG}}(r) = g_{\text{bar,CPTG}}(r) + \Delta g_{\text{pol}}(r) + \Delta g_{\text{geom}}(r), \quad (14)$$

where the leading weak-field polarization behavior is

$$\Delta g_{\text{pol}}(r) \sim \sqrt{a_{\star} g_{\text{bar,CPTG}}(r)}. \quad (15)$$

This square-root response is why CPTG can recover MOND-like weak-field behavior without being a MOND interpolation law. The similarity is phenomenological, not theoretical. MOND modifies the acceleration rule. CPTG derives an added curvature response from baryon-sourced geometric polarization.

In the high-acceleration limit, the polarization contribution is suppressed and the ordinary baryonic limit is recovered:

$$g \gg a_{\star} \implies g_{\text{CPTG}}(r) \approx g_{\text{bar,CPTG}}(r). \quad (16)$$

In the weak-field limit, the polarization response becomes active:

$$g \ll a_{\star} \implies g_{\text{CPTG}}(r) \sim \sqrt{a_{\star} g_{\text{bar,CPTG}}(r)} + g_{\text{geom}}(r). \quad (17)$$

This is the first major distinction between CPTG and dark matter modeling. CPTG does not place unseen mass where the baryonic source appears insufficient. It computes

how curvature polarizes around the baryonic source and tests whether that response reproduces the observed gravitational behavior.

Curvature polarization is therefore the local source-response side of CPTG. It explains why baryonic matter can generate more gravitational support than its Newtonian field alone would imply, without introducing a separate non-baryonic matter component.

7 What Curvature Transport Means

Curvature transport is the organized movement, redistribution, or projection of curvature response across a system or across observational coordinates. It is not the transport of ordinary matter and it is not a new particle fluid. It is the directional or scalable behavior of the curvature field after baryonic structure has generated a polarized response.

Polarization answers the question:

$$\text{How does curvature respond locally to baryonic structure?} \quad (18)$$

Transport answers the next question:

$$\text{How is that organized curvature response carried, displaced, or projected?} \quad (19)$$

The hierarchy is

$$\begin{array}{c} \text{baryonic curvature source} \\ \downarrow \\ \text{curvature polarization} \\ \downarrow \\ \text{curvature transport} \\ \downarrow \\ \text{observed displacement, scaling, or comparison coordinate} \end{array} \quad (20)$$

In bound galaxy systems, transport appears mainly as a reduced radial remnant inside the solved galaxy field. In dissociative cluster mergers, transport remains more explicit because the system is dynamically anisotropic. Gas, galaxies, and lensing structure can separate. CPTG interprets part of this separation as transported curvature support rather than as collisionless dark matter detached from the gas.

A representative reduced merger-plane lensing expression is

$$K_{\text{lens}} = \nabla^2 \Phi - s \nabla \cdot [\Sigma^2 S (S \cdot \nabla) \Phi] . \quad (21)$$

Here K_{lens} is the projected lensing-curvature response, Φ is the reduced potential, s is the reduced transport strength coefficient, Σ is the transport or suppression factor, and S is

the projected transport direction. The first term is the ordinary curvature source. The second term represents directional transport along S . This allows the lensing response to be displaced relative to the collisional gas when the geometry supports coherent transport.

Transport also appears in cosmology-facing work. In that setting, CPTG-native geometric quantities must be expressed in observational coordinates such as CMB likelihood rows, BBN baryon coordinates, BAO ruler ratios, weak-lensing amplitudes, and Hubble comparison projections . The transport factors are not tuning knobs. They are fixed scalable rules derived from the geometric branch.

For example, the geometric-pi comparison branch uses

$$G_T = \frac{p_{ac}}{p_C}, \quad (22)$$

with

$$p_C = \pi, \quad (23)$$

and

$$p_{ac} = 3 - \frac{\pi}{100}. \quad (24)$$

Once this branch is fixed, transport becomes a scaling rule between native CPTG geometry and observational coordinates. The comparison hierarchy is

$$\begin{array}{c} \text{CPTG native geometric branch} \\ \downarrow \\ \text{locked transport factor} \\ \downarrow \\ \text{comparison-coordinate projection} \\ \downarrow \\ \text{reported observational quantity} \end{array} \quad (25)$$

This is why CPTG can bridge into Planck, DESI, BBN, supernova, growth, and lensing interfaces without changing the underlying formulas. Transport is the mechanism that allows one native geometric branch to scale into many observational languages.

Curvature transport is therefore the organizing and projection side of CPTG. Polarization generates the nonlinear curvature response. Transport carries, displaces, or scales that response into the observational form required by the system being tested.

8 Why CPTG Is Not a Redefinition of General Relativity

CPTG is not a rejection or redefinition of General Relativity. It is a geometric extension of the same basic principle: gravity is curvature .

General Relativity identifies gravitation with spacetime geometry. CPTG keeps that foundation. The difference is that CPTG allows curvature to polarize, organize, and transport in regimes where the ordinary reduced description is incomplete. The purpose is not to replace Einstein geometry with a non-geometric force. The purpose is to strengthen the geometric interpretation by giving curvature additional structure before the theory is forced into singular or dark-sector descriptions.

The schematic distinction is

$$\begin{array}{c}
\text{Einstein geometry} \\
\downarrow \\
\text{nonlinear curvature polarization} \\
\downarrow \\
\text{curvature transport and finite closure} \\
\downarrow \\
\text{observed gravitational behavior}
\end{array} \tag{26}$$

In this sense, CPTG treats General Relativity as the geometric foundation, not as the failed theory. The point is that Einstein's equations should not be interpreted as physically failing at a black-hole singularity. A singularity is better understood as a warning that the reduced classical description has been pushed beyond the range where curvature can be treated as an unstructured continuation to infinite density.

CPTG replaces that endpoint with geometric closure [12, 13]. In a high-gradient curvature state, the question is not whether gravity stops being geometry. The question is how geometry polarizes, saturates, transports, or suppresses further collapse when the ordinary coordinate description would otherwise drive the solution toward a singular limit.

A compact way to express the CPTG extension is

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{27}$$

Here $G_{\mu\nu}$ is the Einstein curvature tensor. The additional terms are not dark matter fluids, external substances, or arbitrary correction factors. They represent geometric response: curvature polarization and curvature transport. CPTG therefore strengthens the geometric side of the equation rather than abandoning it.

The limiting behavior is also important. In compact high-acceleration or ordinary weak-curvature regimes where the added curvature response is suppressed or unnecessary, CPTG must reduce back toward the standard baryonic and relativistic limits:

$$\text{ordinary curvature regime} \implies \Pi_{\mu\nu}^{\text{pol}}, \Theta_{\mu\nu}^{\text{trans}} \text{ small or inactive.} \tag{28}$$

In regimes where the observed data require additional gravitational support, displaced lensing structure, finite collapse behavior, or cosmological comparison scaling, CPTG activates the curvature-response structure:

$$\text{organized curvature regime} \implies \Pi_{\mu\nu}^{\text{pol}}, \Theta_{\mu\nu}^{\text{trans}} \text{ observable through projection.} \tag{29}$$

This is why CPTG is not a competitor to the geometric insight of General Relativity. It is a continuation of that insight. It says that curvature itself may contain the missing structure that is usually assigned to dark matter, dark energy, or singular endpoints.

The black-hole singularity is therefore not treated as proof that Einstein’s theory fails. It is treated as evidence that the geometric description needs a finite curvature-closure rule. CPTG supplies such a rule through curvature polarization, curvature transport, and suppressed or saturated high-gradient response.

The distinction can be stated simply:

$$\begin{array}{ccc} \text{GR: gravity is curvature} & & \\ \downarrow & & (30) \\ \text{CPTG: curvature has nonlinear polarization and transport structure} & & \end{array}$$

CPTG therefore does not weaken General Relativity. It extends the geometric program by refusing to let curvature terminate in either an unphysical singularity or an invented dark component when a structured geometric response can be tested instead.

9 CPTG Capabilities from One Geometric Source

The following tables are the central conceptual inventory of this article. The entries are not presented as unrelated modules. They are reduced expressions or scalable projections of the same geometric framework. The structure, lensing, cosmology, abundance, collapse, compact-object, and high-redshift examples are developed in the companion CPTG papers [11, 14, 15, 16].

CPTG capability	Core geometric source	Scalable observational expression
Galaxy rotation support	Baryon-sourced curvature polarization and the solved weak-field response	Rotation curves are computed from the baryonic source without assigning a dark halo or changing the governing formula.
Structural Mode N	Curvature-weighted organization of the solved acceleration field	A theory-derived structural observable is computed from the solved curvature field rather than assigned from morphology, halo concentration, or catalog type.
Outer-slope convergence	Far-outer continuation of the same reduced galaxy equation	Galaxy outskirts are predicted to approach a stable curvature-polarization continuation rather than drifting arbitrarily once the source is fixed.
Dissociative cluster lensing	Curvature polarization plus directional curvature transport	Gas-lensing separation is treated as displaced curvature support, not as a separately inserted collisionless matter component.
Cluster active-gate response	Baryonic loading, support temperature, redshift, and aperture response	Relaxed or coherent cluster apertures are projected through a fixed active curvature-response gate instead of independent halo refitting.

Table 2: CPTG structure and lensing capabilities as scalable expressions of the same curvature-polarization and curvature-transport framework.

CPTG capability	Core geometric source	Scalable observational expression
Geometric-pi comparison branch	Native branch values $p_C = \pi$, $p_{ac} = 3 - \pi/100$, and $G_T = p_{ac}/p_C$	CMB, BBN, supernova, growth, lensing, BAO, and ShapeFit quantities are expressed as comparison coordinates of one fixed branch.
CMB comparison-map closure	Locked geometric-pi CMB branch projected into real map space	Planck and WMAP map products are tested under a common phase scaffold, mask, resolution, and residual rule.
CMB Route B Option 1	Amplitude-level curvature-transport projection $\Phi_\pi(a, k) = \Phi_0(k)C_T(a, k)$	The power-spectrum coordinate follows $P(k) \rightarrow P(k)C_T(a, k)^2$ and is tested through Planck likelihood-coordinate plumbing.
BBN abundance coordinate	Transport-scaled baryon and radiation conversion	The nuclear-abundance layer uses the transported BBN coordinate rather than directly inserting the acoustic/CMB baryon coordinate.
Cosmological lithium solution	Locked mass-seven geometric gate applied to live ${}^7\text{Li}$ and ${}^7\text{Be}$ channels	The lithium problem is treated as a source-network curvature gate rather than a post-processing abundance division.
Hubble-tension bridge	One native geometric branch projected into acoustic/CMB and local luminosity-distance coordinates	The CMB and local values are interpreted as two comparison projections of one native branch rather than two incompatible expansion histories.
DESI and BAO interfaces	Fixed acoustic-ruler and Alcock-Paczynski comparison-coordinate maps	DESI-style ShapeFit, BAO, and ruler-ratio quantities are treated as survey coordinates of the locked geometric branch.
Weak-lensing and growth	Reduced growth and lensing response from the same curvature branch	S_8 , CMB-lensing, and bandpower quantities are distinguished as projected observables rather than collapsed into one fitted lensing parameter.

Table 3: CPTG cosmology, abundance, and survey-coordinate capabilities as scalable projections of the locked geometric branch.

CPTG capability	Core geometric source	Scalable observational expression
Vanishing-star channels	Structural collapse and transport suppression in high-gradient curvature states	Failed-supernova or vanishing-source behavior is interpreted through finite-core curvature closure and suppressed transport.
Low mass-gap structure	Compact-object finite-core closure and apparent polarization amplification	Compact objects are classified by physical core mass and curvature-gradient state, allowing apparent mass to differ from physical core mass.
High-redshift compact galaxies	Redshift-normalized support scale, curvature radius, and structural stress	Early compact systems are tested as geometric stress cases rather than as isolated stellar-population or halo-tuning anomalies.

Table 4: CPTG collapse, compact-object, and high-redshift stress-test capabilities as derived expressions of the same scalable geometric framework.

10 Structural Mode N and the Curvature Structure Mode Index

Structural Mode N is one of the clearest examples of what makes CPTG different from halo fitting and acceleration-law models [10]. It is not a mass component, not a fitted halo concentration, not a catalog morphology, and not a MOND-like interpolation parameter. It is a post-solve structural diagnostic extracted from the curvature organization of the solved CPTG acceleration field.

After the CPTG galaxy equation has been solved, the reduced curvature-structure invariant is

$$C(r) = g^2(r) + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2. \quad (31)$$

The corresponding curvature-structure weight is

$$w(r) = \left| \frac{dC}{dr} \right|. \quad (32)$$

The outer resolved radius is

$$R = \max(r), \quad (33)$$

and the characteristic curvature-organization length is

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}. \quad (34)$$

The coefficient κ_{struct} carries the structural length weighting needed to compare the field strength and field gradient pieces. The interval $R/5 \leq r \leq R$ restricts the diagnostic to the resolved outer structural field rather than the innermost source dominated region.

The structural mode is then

$$N = \frac{R}{\lambda}. \quad (35)$$

The same quantity is reported in the benchmark workflow as the Curvature Structure Mode Index, or CSMI. The CSMI is the continuous numerical value of N , while the associated CSMI type is the structural family label assigned from that value. This makes the mode type a theory-derived classification: the galaxy is first solved, the curvature organization is measured, and only then is a structural label assigned.

The logical order is

$$\begin{array}{c}
\text{observed baryonic source} \\
\downarrow \\
\text{solve the CPTG acceleration field} \\
\downarrow \\
\text{compute the CSMI value } N \\
\downarrow \\
\text{assign the CSMI structural type}
\end{array} \tag{36}$$

This is very different from ordinary morphology. A CSMI type is not assigned from the galaxy name, Hubble type, visual appearance, color, surface brightness label, or catalog classification. It is derived from the solved curvature field. In that sense, CSMI is a geometric classification system: it asks what structural type the galaxy has according to the curvature response computed by CPTG.

The three worked examples illustrate how the same diagnostic spans different galaxy regimes without changing the calculation procedure.

Galaxy	CSMI N	CSMI type	Structural interpretation
DDO-154	1.890702489	LSB Dwarf Disk	Low-mode dwarf-disk curvature organization in a gas-rich, low-acceleration system.
NGC-3198	3.062572658	Early Spiral $\leftarrow$ Intermediate Spiral	Extended disk system whose solved curvature mode lies near the intermediate-to-early spiral structural transition.
NGC-7814	3.157080935	Early Spiral	Bulge-dominated system whose solved field remains in the early-spiral CSMI family.

Table 5: Example CSMI classifications from the three worked CPTG rotation-curve systems. The CSMI type is derived from the solved structural mode N , not assigned from catalog morphology.

These examples show why Structural Mode N is not merely a residual statistic. The rotation curve tests whether the solved acceleration field matches the observed velocity profile. The CSMI asks a second question: does the solved curvature field organize itself like the structural family the galaxy physically resembles?

That gives CPTG a capability not present in ordinary dark-halo fitting. A dark-halo model can assign a halo profile after the fact. CPTG computes a curvature-organization index from the solved baryon-sourced field itself. If the same reduced galaxy equation produces the rotation curve and also returns a coherent CSMI classification, the result is stronger than a velocity fit alone.

The CSMI therefore functions as a structural readout of the theory:

$$\begin{array}{cc}
\text{fit quality} & \text{CSMI} \\
\downarrow & \downarrow \\
\text{observed velocity field} & \text{curvature-organization type}
\end{array} \tag{37}$$

This makes N falsifiable. If improved data showed that CSMI values scatter randomly, fail to correlate with structural behavior, or assign incoherent mode families across well-measured galaxies, the structural interpretation would fail. If the mode families remain coherent across different databases and galaxy classes, then CPTG has not only fit rotation curves; it has extracted a physical classification from the geometry of the solved gravitational field.

11 Clarifying the Difference Between CPTG and MOND

CPTG can produce MOND-like weak-field behavior in some galaxy regimes, but CPTG is not a MOND model [1]. The similarity is phenomenological. The theories use different source fields, different acceleration scales, different equations, and different physical interpretations.

MOND begins with a conventional baryonic source and applies a universal acceleration scale. In the benchmark comparison used here, the MOND side uses the same observed SPARC component data [3], but builds the baryonic source with unit weights. The signed-square convention used for component fields is

$$Q(V) = V|V|, \quad (38)$$

which reduces to V^2 for positive component velocities and preserves source orientation when a tabulated component carries a sign. The MOND source is then

$$V_{\text{bar,MOND}}^2(r) = Q(V_{\text{gas}}(r)) + Q(V_{\text{disk}}(r)) + Q(V_{\text{bulge}}(r)). \quad (39)$$

The corresponding MOND baryonic acceleration is

$$g_{\text{bar,MOND}}(r) = \frac{V_{\text{bar,MOND}}^2(r)}{r}. \quad (40)$$

The MOND comparison then applies a fixed universal acceleration scale,

$$a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}, \quad (41)$$

through the benchmark acceleration law

$$g_{\text{MOND}}(r) = \frac{g_{\text{bar,MOND}}(r)}{1 - \exp \left[-\sqrt{g_{\text{bar,MOND}}(r)/a_0} \right]}. \quad (42)$$

This fixed low acceleration scale explains why MOND tends to perform best in low-acceleration systems. Its transition scale is already placed near the weak-field regime where dwarf and low-surface-brightness galaxies live. In those systems, the MOND enhancement is naturally active. As the structure becomes more centrally concentrated, higher acceleration, or more internally organized, the same fixed a_0 must still be used.

MOND cannot recompute a structural scale from the solved system because a_0 is universal by construction.

The MOND logic is therefore

$$\begin{array}{c}
\text{standard baryonic source} \\
\downarrow \\
\text{fixed universal acceleration scale } a_0 \\
\downarrow \\
\text{MOND acceleration law} \\
\downarrow \\
\text{predicted rotation curve}
\end{array} \tag{43}$$

CPTG uses the same observed galaxy data, but it does not use the MOND source field, the MOND acceleration scale, or the MOND law. The CPTG source is constructed with fixed geometric component weights:

$$V_{\text{bar,CPTG}}^2(r) = \frac{27}{50}Q(V_{\text{gas}}(r)) + \frac{23}{50}Q(V_{\text{disk}}(r)) + \frac{16}{25}Q(V_{\text{bulge}}(r)). \tag{44}$$

The CPTG baryonic acceleration is

$$g_{\text{bar,CPTG}}(r) = \frac{V_{\text{bar,CPTG}}^2(r)}{r}. \tag{45}$$

The key difference is the acceleration scale. MOND uses a fixed universal scale a_0 . CPTG uses a structure-specific scale $a_\star$. This scale belongs to the baryonic source structure, curvature-polarization response, and transport branch being evaluated. It is not a universal MOND-like constant and it is not carried from one structure to another.

The structural rule is

$$a_\star \rightarrow a_\star[\text{structure, baryonicsource, transportbranch}]. \tag{46}$$

For a galaxy rotation-curve calculation, the structure is the resolved galaxy. For another bound system, the relevant structure must be evaluated on its own terms. The exception is the universe-scale branch, where CPTG uses the native geometric acceleration anchor $a_{\pi,0}$ rather than a locally inferred $a_\star$.

Once $a_\star$ is determined for the structure, the transport radius is derived directly:

$$R_c = \frac{48\pi c^2}{a_\star}. \tag{47}$$

Thus R_c is not an independent halo radius and not a second free scale. It is the transport scale implied by the structural acceleration scale.

The reduced CPTG galaxy equation is then solved implicitly:

$$g_{\text{CPTG}}(r) = g_{\text{bar,CPTG}}(r) + \frac{\sqrt{a_\star g_{\text{bar,CPTG}}(r)} + g_{\text{geom}}(r)}{\left[1 + (g(r)/a_\star)^{123/50}\right]^{11/50}}. \quad (48)$$

Here $g(r)$ denotes the same solved total acceleration field, $g_{\text{CPTG}}(r)$, because the equation is implicit in the total response.

The geometric transport remnant is

$$g_{\text{geom}}(r) = \eta a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad (49)$$

with the fixed reduced transport coefficient

$$\eta = \frac{1}{2}. \quad (50)$$

The CPTG logic is therefore

$$\begin{array}{c} \text{CPTG-weighted baryonic source} \\ \downarrow \\ \text{structure-specific scale } a_\star \\ \downarrow \\ \text{derived transport radius } R_c \\ \downarrow \\ \text{implicit curvature-polarization solve} \\ \downarrow \\ \text{predicted rotation curve and CSMI} \end{array} \quad (51)$$

The benchmark behavior follows directly from this distinction. MOND has its best natural leverage in the low-acceleration regime because its fixed a_0 is low. CPTG does not rely on one universal acceleration threshold. It recomputes the structural scale for the solved system, then derives the corresponding transport scale and curvature response. This is why CPTG can operate across dwarf, low-surface-brightness, late-spiral, early-spiral, bulged, and high-mode systems without changing the governing equation.

The operational distinction can be summarized as follows.

Feature	MOND comparison	CPTG galaxy reduction
Baryonic source	Unit-weight gas, disk, and bulge source	Geometrically weighted CPTG source using 27/50, 23/50, and 16/25
Acceleration scale	Fixed universal scale a_0 , positioned in the low-acceleration regime	Structure-specific scale a_* ; universe-scale branch uses $a_{\pi,0}$
Best natural regime	Low-acceleration systems where the fixed a_0 transition is already active	Any solved structure whose baryonic source and curvature organization define a consistent a_*
Main equation	Acceleration law applied to $g_{\text{bar},\text{MOND}}$	Implicit nonlinear curvature-polarization equation solved for g_{CPTG}
Transport radius	Not used	$R_c = 48\pi c^2/a_*$, derived from the structural scale
Structural output	Rotation curve only	Rotation curve plus post-solve CSMI structural mode
Physical interpretation	Modified acceleration behavior with a universal transition scale	Baryon-sourced curvature polarization and transport response of the solved structure

Table 6: Operational differences between the MOND comparison and the CPTG galaxy reduction.

This is why CPTG should not be described as MOND with different constants. MOND applies a universal acceleration law to a standard baryonic source. CPTG constructs a geometric source field, determines the structural scale for the system being solved, derives the transport scale from that structural scale, and solves a nonlinear curvature-response equation.

The weak-field CPTG behavior,

$$g_{\text{CPTG}}(r) \sim \sqrt{a_* g_{\text{bar},\text{CPTG}}(r)} + g_{\text{geom}}(r), \quad (52)$$

can resemble MOND-like phenomenology, but its origin is different. In MOND, the enhancement comes from a modified acceleration law controlled by a_0 . In CPTG, the enhancement is the reduced expression of baryon-sourced curvature polarization controlled by the structure-specific a_* .

The final distinction is compact:

$$\begin{array}{cc}
\text{MOND} & \text{CPTG} \\
\downarrow & \downarrow \\
\text{fixed } a_0 \text{ law} & \text{structure-specific } a_* \text{ solve} \\
\downarrow & \downarrow \\
\text{rotation curve} & \text{rotation curve plus CSMI}
\end{array} \quad (53)$$

12 Cluster Active Gate Response

The cluster active gate is one of CPTG's strongest structure scale results. It is not a galaxy rotation formula applied to clusters, and it is not a dark halo reconstruction. A galaxy rotation curve is a resolved radial acceleration problem. A galaxy cluster is an aperture resolved baryonic and thermal system. CPTG therefore treats clusters through declared aperture inputs: redshift, overdensity radius, baryonic loading, support temperature, and the active curvature state of that aperture.

The practical input layer is

$$\begin{array}{c} z, \Delta, R_{\Delta}, M_b(< R_{\Delta}), T_{\text{gate}} \\ \downarrow \\ F_{\Delta}, Q_C, y_{\Delta}, N_{C,\Delta}, M_{\text{CPTG},\Delta}. \end{array} \quad (54)$$

The key rule is aperture matching. The baryonic mass, gas temperature, predicted CPTG mass, and comparison mass must all refer to the same declared aperture. This prevents the calculation from mixing inner gas information, outer temperature information, and mass estimates from different radii.

The active gate variable Q_C classifies the aperture response. High Q_C apertures approach closure. Low Q_C apertures enter suppressed response states. The solved response ratio y_{Δ} then gives the predicted cluster mass,

$$M_{\text{CPTG},\Delta} = y_{\Delta} M_{\Delta,\text{ref}}. \quad (55)$$

Here $M_{\Delta,\text{ref}}$ is the branch native reference mass at the declared aperture. The active response ratio y_{Δ} converts that reference mass into the predicted CPTG aperture mass.

The cluster structural mode is

$$N_{C,\Delta} = \frac{y_{\Delta} N_{C,0}}{Q_C^2}. \quad (56)$$

This quantity is not a mass multiplier. It is a response load variable. It measures how deeply the selected aperture sits inside the active curvature response branch. Suppressed apertures carry higher response load. Closure stable apertures carry lower response load. The full derivation of F_{Δ} , θ_T , θ_{crit} , Q_C , y_{Δ} , and $N_{C,\Delta}$ is given in the dedicated cluster active gate paper [9].

This is where CPTG differs sharply from competing approaches. In a Λ CDM style cluster treatment, the missing support is assigned to a dark halo and then organized through halo profile, concentration, relaxation state, hydrostatic bias, simulation calibration, and aperture convention. In a MOND style treatment, the galaxy scale acceleration idea does not by itself provide a baryonic loading and temperature supported aperture gate

for cluster masses. Hydrostatic reconstruction is useful as an observational comparison product, but it is not a geometric explanation of why the aperture closes.

CPTG instead computes the cluster aperture state from the declared baryonic and thermal inputs. The distinction is

dark halo	MOND style	CPTG
↓	↓	↓
assign halo support	apply acceleration law	compute active aperture
↓	↓	↓
profile and bias dependence	cluster residual stress	baryon temperature closure

(57)

The tested performance is the important point. In the X-COP same aperture R_{500} comparison, the active gate was tested against 12 hydrostatic mass cluster apertures. The result was

$$\text{median} \left(\frac{M_{\text{CPTG},500}}{M_{\text{HSE},500}} \right) = 0.9892, \quad (58)$$

with

$$\text{median} \left| \frac{M_{\text{CPTG},500}}{M_{\text{HSE},500}} - 1 \right| \simeq 0.0108, \quad (59)$$

and

$$\max \left| \frac{M_{\text{CPTG},500}}{M_{\text{HSE},500}} - 1 \right| \simeq 0.0268. \quad (60)$$

This is percent level same aperture agreement in a regime where cluster mass modeling is normally sensitive to halo assumptions, gas support, hydrostatic bias, relaxation state, and aperture definition.

The ACCEPT profile diagnostic gives a separate response ordering check. Closure stable rows have the largest median Q_C and remain near $y_\Delta = 1$. Suppressed rows have lower median Q_C and lower y_Δ . Strongly suppressed rows sit at the lowest median Q_C and show the strongest reduction in y_Δ . The active gate therefore sorts cluster profile rows by curvature response state rather than merely returning a fitted mass number.

The X-COP overdensity ladder adds the aperture structure. The median gate decreases coherently from the inner apertures toward R_{500} ,

$$\begin{aligned}
Q_C(R_{2500}) &> Q_C(R_{2000}) > Q_C(R_{1500}) \\
&> Q_C(R_{1000}) > Q_C(R_{750}) > Q_C(R_{500}).
\end{aligned}
\tag{61}$$

In the tested ladder, the median gate moves from approximately

$$Q_C(R_{2500}) \simeq 1.5781 \tag{62}$$

to

$$Q_C(R_{500}) \simeq 0.9701. \tag{63}$$

The inner apertures sit closer to closure. The outer apertures expose the active gate transition. This behavior is not a halo fit and not a low acceleration interpolation. It is an aperture based curvature response law.

The cluster active gate is therefore a distinctive CPTG capability. It connects baryonic loading, support temperature, redshift, aperture radius, predicted mass, gate class, and structural response load in one geometric calculation. In the tested X-COP regime, that calculation reaches percent level same aperture closure without assigning a dark halo and without importing a galaxy scale MOND law.

13 Validation by Refusal to Tune

The most important methodological difference is that CPTG does not gain strength by adding freedom . It gains strength by refusing to add freedom after the branch is locked.

The validation rule is

$$\begin{aligned}
&\text{derive the formula} \\
&\quad \downarrow \\
&\text{lock the formula} \\
&\quad \downarrow \\
&\text{scale the formula} \\
&\quad \downarrow \\
&\text{test the projection}
\end{aligned}
\tag{64}$$

If the projection fails, the branch is stressed. If it survives, the result matters because the formula did not move.

This is why CPTG should not be judged by whether it can reproduce every step of a tuned dark-sector pipeline. It should be judged by whether its fixed geometric branch can reproduce the observational structure that the tuned pipeline was built to explain.

14 Falsifiability

CPTG is falsifiable because the formulas are not allowed to move freely. The theory would be stressed if any clean data sector required changing the native branch, changing the transport map, adding a sector-specific geometric patch, inflating lensing amplitude by hand, replacing the lithium gate, or assigning a hidden halo to rescue a solved curvature response.

Examples of direct stress tests include:

$$\begin{aligned} &\text{mode-selected galaxies lack structural coherence;} \\ &\text{active-gate apertures fail matched baryonic inputs;} \\ &\text{CMB coordinates require lensing inflation or posterior retuning;} \\ &\text{BBN controls reject the transported baryon coordinate;} \\ &\text{future DESI, CMB, or galaxy data reject the locked projection.} \end{aligned} \tag{65}$$

These are not vague philosophical risks. They are concrete failure modes. CPTG's strength comes from the fact that the same locked structure has to keep surviving.

15 Conclusion

CPTG is different because it does not begin with missing matter, missing energy, or a new interpolation rule. It begins with geometry. Ordinary baryonic structure sources curvature polarization, curvature transport, active response, and comparison coordinate projection. Those formulas are derived from the core theory, locked before testing, and then scaled into the observational coordinate required by the data.

This is the central distinction. A dark matter framework can add, reshape, displace, suppress, or redistribute an unseen halo when a structure does not behave as expected. A dark energy framework can absorb expansion behavior into a fitted background component. A MOND style framework can improve galaxy scale behavior by using a fixed low acceleration scale, but it does not naturally extend into cluster aperture closure, CMB comparison coordinates, BBN transport, or source network lithium gates.

CPTG takes the opposite route. It does not ask what invisible component must be placed into the system. It asks whether the observed behavior is a scalable expression of the same geometric source. The calculation either closes or it does not; the formula either projects correctly or it fails. That refusal to retune is what makes the framework testable.

The result is a level of explanatory compression that competing theories do not currently match. Structural Mode N and CSMI classify galaxies from the solved curvature field rather than from catalog morphology. The cluster active gate predicts same aperture cluster response from baryonic loading and thermal support rather than from halo assignment. The geometric π branch scales into CMB, BBN, BAO, DESI, Hubble, growth, and lensing comparison coordinates. The lithium gate acts inside live source network channels rather than dividing the abundance after the fact. Route B and the CMB map

closure tests show that CPTG can enter Planck facing data spaces as a fixed geometric projection rather than as a movable perturbation fit.

The important point is not that every observational interface looks the same. A galaxy rotation curve, a cluster aperture, a CMB likelihood coordinate, a BBN abundance row, a DESI ruler coordinate, a lithium source channel, and a high redshift stress test are different data products. The important point is that CPTG does not invent a separate theory for each one. The interfaces differ, but the origin does not.

The validation standard is therefore

$$\begin{array}{c}
 \text{derive the geometry} \\
 \downarrow \\
 \text{lock the formula} \\
 \downarrow \\
 \text{scale into the observed coordinate} \\
 \downarrow \\
 \text{test without sector by sector retuning}
 \end{array}
 \tag{66}$$

This is a harder standard than ordinary parameter fitting. A theory with enough adjustable structure can often be made to follow a particular dataset. CPTG is testing whether a compact set of geometric rules can keep surviving independent regimes without being moved to meet them.

That is why the fragility of dark matter is relevant. The observations are real, but the interpretation remains operationally fragile whenever the unseen sector must be rearranged to rescue a structure that does not behave as expected. CPTG does not need that rescue mechanism. It lets the baryonic source, curvature response, transport branch, active gate, or comparison coordinate carry the prediction.

The strongest CPTG claim is therefore not that it is more adjustable than competing models. It is that it is less adjustable and still works across more regimes than a fixed geometric theory would be expected to survive.

The framework can be summarized in one line:

$$\begin{array}{c}
 \text{one geometric source} \\
 \downarrow \\
 \text{many scalable observational projections} \\
 \downarrow \\
 \text{no sector by sector tuning}
 \end{array}
 \tag{67}$$

If future data reject the locked projections, CPTG fails in the open. If the same locked geometry continues to close across galaxies, clusters, CMB, BBN, lithium, Hubble, DESI, growth, compact objects, and high redshift structures, then CPTG is not merely an

alternative explanation. It is evidence that the missing structure in modern cosmology may have been geometric all along.

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